NT-1: Literature Extraction & Convention Fixing for SCT Nonlocal Form Factor Computation

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Abstract

This document collects all formulas, conventions, and computational recipes needed to explicitly derive the nonlocal form factors $F_1(\Box/\Lambda^2)$ and $F_2(\Box/\Lambda^2)$ from the SCT spectral action $\mathrm{Tr}(f(D^2/\Lambda^2))$. All formulas are extracted from the literature; no original derivations are performed. Each formula is annotated with its source, equation number, and whether it matches our sign conventions. Gaps are noted explicitly.

Key references:

- Barvinsky–Vilkovisky (BV): Nucl. Phys. B **282** (1987) 163; B **333** (1990) 471, 512; arXiv:0911.1168.
- Avramidi: Phys. Lett. B **236** (1990) 443; B **238** (1990) 92; hep-th/9704166.
- Codello–Zanusso (CZ): arXiv:1203.2034.
- Vassilevich: Phys. Rept. **388** (2003) 279 [hep-th/0306138].
- Iochum–Levy–Vassilevich (ILV): arXiv:1108.3749; 1201.6637.

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1 Conventions

All conventions are fixed here and **must be used consistently** throughout the NT-1 computation. Where references differ, we note the conversion.

1.1 Spacetime and metric

Convention 1.1 (Metric signature)**.** [our convention] We use the **mostly-plus** Lorentzian signature:

$$\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)\,. \tag{1}$$

For the heat kernel computation we Wick-rotate to **Euclidean signature** $(+, +, +, +)$. The Euclidean metric is denoted $g_{\mu\nu}$ (positive definite).

Remark 1.2. Vassilevich [13], Codello–Zanusso [14], and the BV series [2, 3] all work in Euclidean signature. No sign adjustment needed for formulas extracted from these sources (they are already Euclidean). The Wick rotation from our Lorentzian convention is $t \to -\mathrm{i}\tau$, $g_L^{00} = -1 \to g_E^{00} = +1$.

Convention 1.3 (Spacetime dimension)**.** $d = 4$ for all final results. Intermediate formulas are kept general in d where possible.

1.2 Curvature tensors

Convention 1.4 (Riemann tensor)**.** [our convention]

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}\,. \tag{2}$$

This is the "$+--+$" convention in the MTW classification (same as Wald, Vassilevich, Avramidi).

Convention 1.5 (Ricci tensor)**.** [our convention] Contraction on **first and third** indices:

$$R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}\,. \tag{3}$$

This matches Vassilevich [13] and Codello–Zanusso [14] (who write $R_{\mu\nu} \equiv R_{\alpha\mu}{}^{\alpha}{}_{\nu}$, which is the same thing).

Convention 1.6 (Scalar curvature)**.**

$$R = g^{\mu\nu} R_{\mu\nu}\,. \tag{4}$$

Positive for a round sphere in Euclidean signature.

Convention 1.7 (Weyl tensor)**.** In d dimensions:

$$C_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} - \frac{2}{d-2}\Big(g_{\mu[\rho}R_{\sigma]\nu} - g_{\nu[\rho}R_{\sigma]\mu}\Big) + \frac{2}{(d-1)(d-2)}\,R\,g_{\mu[\rho}g_{\sigma]\nu}\,. \tag{5}$$

Its square satisfies [CZ (A.3)]:

$$C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - \frac{4}{d-2}R_{\mu\nu}R^{\mu\nu} + \frac{2}{(d-1)(d-2)}\,R^2\,. \tag{6}$$

In $d = 4$: $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 2R_{\mu\nu}R^{\mu\nu} + \frac{1}{3}R^2$.

Convention 1.8 (Gauss–Bonnet / Euler density)**.**

$$E_4 = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 4R_{\mu\nu}R^{\mu\nu} + R^2\,. \tag{7}$$

Topological in $d = 4$: $\int E_4\sqrt{g}\,\mathrm{d}^4x = 32\pi^2\chi(M)$.

1.3 Differential operators

Convention 1.9 (Covariant derivative and d'Alembertian)**.** [our convention] The covariant d'Alembertian (Laplace–Beltrami operator) on scalars:

$$\Box \equiv g^{\mu\nu}\nabla_\mu\nabla_\nu\,. \tag{8}$$

In Lorentzian signature this is the wave operator ($\Box\phi = -\partial_t^2\phi + \nabla^2\phi$ in flat space). In **Euclidean signature** this becomes:

$$\Box_E = g^{\mu\nu}\nabla_\mu\nabla_\nu = \nabla^2 \quad \text{(positive spectrum on compact manifold)}\,. \tag{9}$$

Convention 1.10 (Generalised Laplacian)**.** [our convention] A Laplace-type operator on sections of a vector bundle:

$$\boxed{\Delta = -(g^{\mu\nu}D_\mu D_\nu + E) = -D^2 - E} \tag{10}$$

where $D_\mu = \nabla_\mu + A_\mu$ is the full covariant derivative (including gauge/spin connection) and E is the endomorphism (potential term).

This matches Vassilevich [13] Eq. (1.1) and Codello–Zanusso [14] Eq. (2).

Sign note: CZ define $\Delta = -D^2 + \mathbf{U}$ with $\mathbf{U}$ their endomorphism. Comparing with our convention $\Delta = -D^2 - E$, we identify:

$$\mathbf{U}_{\mathrm{CZ}} = -E\,. \tag{11}$$

This sign difference is critical when translating form factors.

Convention 1.11 (Bundle curvature)**.** The curvature of the connection A_μ on the vector bundle:

$$\Omega_{\mu\nu} \equiv [D_\mu, D_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]\,. \tag{12}$$

This matches both CZ Eq. (1) and Vassilevich.

Convention 1.12 (BV endomorphism $\hat{P}$)**.** The BV papers use $\hat{P} \equiv E + \frac{R}{6}\mathrm{Id}$ (equivalently $\hat{P} = -\mathbf{U}_{\mathrm{CZ}} + \frac{R}{6}\mathrm{Id}$) [Vassilevich Eq. (4.3); CZ App. A]. In our conventions for the Dirac operator:

$$E = -\frac{R}{4}\mathrm{Id}_4 \quad \Longrightarrow \quad \hat{P} = -\frac{R}{4}\mathrm{Id}_4 + \frac{R}{6}\mathrm{Id}_4 = -\frac{R}{12}\mathrm{Id}_4\,. \tag{13}$$

The CZ relation is $\mathbf{U}_{\mathrm{CZ}} = -\hat{P} + \frac{R}{6}\mathrm{Id} = -E$, so $\hat{P} = -\mathbf{U}_{\mathrm{CZ}} + \frac{R}{6}\mathrm{Id}$ [CZ App. A].

[Sign correction] Convention 2.11 corrected 2026-03-09: the BV endomorphism is $\hat{P} = E + R/6$ (not $E - R/6$ as stated in the original version). The original error gave $\hat{P} = -5R/12$ instead of the correct $\hat{P} = -R/12$ for Dirac. This error was *contained*: the NT-1 derivation pipeline (Proposition 3.2, Theorem 4.1 of `NT1_form_factors.tex`) works entirely through CZ form factors and never uses $\hat{P}$ in the computational chain. See the failed derivation in Appendix A of `NT1_form_factors.tex` for the consequences of the wrong $\hat{P}$.

Remark 1.13 ($\hat{P}$-independence of NT-1)*.* The NT-1 derivation of the nonlocal form factors F_1, F_2 is *structurally independent* of the BV endomorphism $\hat{P}$. The computational pipeline proceeds as follows:

1. Extract operator data $(E, \Omega_{\mu\nu})$ from the Lichnerowicz formula (Convention 15).

2. Map to the CZ universal form factors $f_{\mathrm{Ric}}, f_R, f_{R\mathbf{U}}, f_{\mathbf{U}}, f_\Omega$ using the CZ notation $\mathbf{U}_{\mathrm{CZ}} = -E$ (Convention 2.10).
3. Contract CZ form factors with Dirac traces $\mathrm{tr}(\mathrm{Id})$, $\mathrm{tr}(E)$, $\mathrm{tr}(E^2)$, $\mathrm{tr}(\Omega_{\mu\nu}\Omega^{\mu\nu})$ to obtain $h_C(x)$ and $h_R(x)$.

At no point does $\hat{P}$ or its traces appear in this chain. The BV form factors $f_1, \ldots, f_5$ (which do involve $\hat{P}$) are used only in §3 of `NT1_form_factors.tex` as an alternative cross-check pathway, which was found to be algebraically more cumbersome and is not part of the main derivation.

1.4 Dirac operator specifics

Convention 1.14 (SCT Dirac operator)**.** On a closed Riemannian spin 4-manifold (M, g):

$$D = \mathrm{i}\gamma^\mu \nabla^S_\mu \,, \tag{14}$$

where $\nabla^S_\mu = \nabla_\mu + \frac{1}{4}\omega^{ab}_\mu \gamma_a \gamma_b$ is the spin connection covariant derivative.

Convention 1.15 (Lichnerowicz formula)**.**

$$D^2 = -g^{\mu\nu}\nabla^S_\mu \nabla^S_\nu + \frac{R}{4} \,. \tag{15}$$

So D^2 is of Laplace type (10) with:

$$\boxed{E = -\frac{R}{4}\mathrm{Id}_4 \,, \qquad \Omega^{(\mathrm{spin})}_{\mu\nu} = \frac{1}{4} R_{\mu\nu\rho\sigma}\gamma^\rho\gamma^\sigma} \tag{16}$$

and $\mathbf{U}_{\mathrm{CZ}} = -E = \frac{R}{4}\mathrm{Id}_4$.

Remark 1.16 (**Sign of E — critical**). The Lichnerowicz formula gives $D^2 = -g^{\mu\nu}\nabla\nabla + R/4$. Comparing with the generalized Laplacian $\Delta = -(g^{\mu\nu}\nabla\nabla + E)$, we identify $E = -R/4$ (**not** $+R/4$). Cross-check: $E = -R/4$ gives $h_R(0) = 0$, confirming conformal invariance of the massless Dirac field. The wrong sign $E = +R/4$ would give $h_R(0) \neq 0$.

Convention 1.17 (Spin connection curvature trace identities)**.** [exact from lit.]

$$\begin{aligned} \mathrm{tr}(\Omega^{(\mathrm{spin})}_{\mu\nu}\Omega^{(\mathrm{spin})\mu\nu}) &= \frac{1}{16} R_{\mu\nu\rho\sigma}R_{\alpha\beta\gamma\delta}\ \mathrm{tr}(\gamma^\rho\gamma^\sigma\gamma^\gamma\gamma^\delta)\, g^{\mu\alpha}g^{\nu\beta} \\ &= -\frac{1}{2} R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \,. \end{aligned} \tag{17}$$

(The factor $N_{\mathrm{spin}} = \mathrm{tr}\,\mathrm{Id}_4 = 4$ in $d = 4$.)

1.5 Heat kernel and spectral action

Convention 1.18 (Heat kernel)**.**

$$K(s; x, y) = \langle x|e^{-s\Delta}|y\rangle \,, \qquad \mathrm{Tr}\ K(s) = \mathrm{Tr}\ e^{-s\Delta} \,. \tag{18}$$

Here $s > 0$ is the proper-time parameter. The heat kernel satisfies $(\partial_s + \Delta_x)K(s; x, y) = 0$ with $K(0; x, y) = \delta(x, y)$. This matches CZ (10)–[14] Eqs. (3)–(4).

Convention 1.19 (SCT spectral action and heat kernel)**.** The spectral action is

$$S_{\mathrm{SCT}} = \mathrm{Tr}\Big(f(D^2/\Lambda^2)\Big)\,. \tag{19}$$

Using the Laplace transform $f(u) = \int_0^\infty \tilde{f}(t)\, e^{-tu}\mathrm{d}t$:

$$\mathrm{Tr}\Big(f(D^2/\Lambda^2)\Big) = \int_0^\infty \tilde{f}(t)\ \mathrm{Tr}\Big(e^{-tD^2/\Lambda^2}\Big)\mathrm{d}t\,. \tag{20}$$

With $s = t/\Lambda^2$:

$$\boxed{\mathrm{Tr}\Big(f(D^2/\Lambda^2)\Big) = \Lambda^2 \int_0^\infty \tilde{f}(s\Lambda^2)\ \mathrm{Tr}\Big(e^{-sD^2}\Big)\mathrm{d}s} \tag{21}$$

Note $D^2 = \Delta$ (our Laplace-type operator for the Dirac case).

Convention 1.20 (Relation to one-loop effective action)**.** The standard one-loop effective action for a field with operator Δ and mass m:

$$\Gamma^{(1)} = \mp\frac{1}{2}\ln\det\Delta = \pm\frac{1}{2}\int_0^\infty \frac{\mathrm{d}s}{s}\, e^{-sm^2}\ \mathrm{Tr}\ e^{-s\Delta}\,, \tag{22}$$

where $-$ for bosons, $+$ for fermions (Dirac). Compare with (21): the spectral action replaces the $1/s$ weight with a smooth cutoff function.

2 Avramidi Covariant Technique for Off-Diagonal Heat Kernel

2.1 The problem: local vs. nonlocal

The Seeley–DeWitt expansion gives the heat trace as a series in powers of proper time s:

$$\mathrm{Tr}\, e^{-s\Delta} \sim \frac{1}{(4\pi s)^{d/2}} \sum_{k=0}^{\infty} s^k\, a_{2k}(\Delta)\,, \qquad s \to 0^+\,. \tag{23}$$

This is **local**: each a_{2k} is a spacetime integral of local curvature invariants. It captures only the **coincidence limit** $x = y$ of the heat kernel diagonal.

To obtain the **nonlocal form factors** $F_1(\Box/\Lambda^2)$, $F_2(\Box/\Lambda^2)$ we need the heat kernel expanded to second order in curvature but to **all orders in derivatives** (i.e., all orders in s, not just the $s \to 0$ asymptotics).

2.2 Avramidi's partial summation

[Avramidi, Phys. Lett. B **236** (1990) 443; Phys. Lett. B **238** (1990) 92; hep-th/9704166]

Avramidi's key insight: decompose the heat kernel expansion not by powers of s (local) but by **powers of curvature**, while keeping all powers of ∇ (i.e., all powers of $s\Box$). This "partial summation" resumes the Seeley–DeWitt series into nonlocal form factors.

Definition 2.1 (Avramidi form factor basis)**.** At second order in curvature, the nonlocal heat trace on a boundaryless manifold takes the form [Avramidi (1990); CZ Eq. (23)]:

$$\boxed{\begin{aligned}\mathrm{Tr}\, e^{-s\Delta} = \frac{1}{(4\pi s)^{d/2}} \int \mathrm{d}^d x\, \sqrt{g}\, \mathrm{tr}\Big\{ &\mathrm{Id} + s\hat{P} + s^2\Big[\mathrm{Id}\, R_{\mu\nu}\, f_1(s\Box)\, R^{\mu\nu} \\ &+ \mathrm{Id}\, R\, f_2(s\Box)\, R + R\, f_3(s\Box)\, \hat{P} + \hat{P}\, f_4(s\Box)\, \hat{P} + \Omega_{\mu\nu}\, f_5(s\Box)\, \Omega^{\mu\nu}\Big] + \mathcal{O}(\mathcal{R}^3)\Big\}\end{aligned}} \tag{24}$$

where $\hat{P} = E - \frac{R}{6}\mathrm{Id}$ (Convention 1.12), $\Box = -D^2$ (**without** the endomorphism), and f_i are the five BV form factors.

Remark 2.2. This expansion is valid under the **covariant perturbation theory condition** [CZ (24)]:

$$\nabla\nabla\mathcal{R} \gg \mathcal{R}^2\,, \tag{25}$$

where $\mathcal{R}$ represents any curvature. Physically: curvatures are weak but may vary rapidly. This is complementary to the Seeley–DeWitt regime (strong/slowly varying fields).

2.3 The basic form factor (master function)

Definition 2.3 (Master form factor)**.** [exact from lit.] [CZ Eq. (36); Avramidi (1990); BV (1987)]

$$\boxed{f(x) = \int_0^1 \mathrm{d}\xi\; e^{-\xi(1-\xi)x}} \tag{26}$$

where $x = s\Box$ (or $x = sp^2$ in momentum space).

Properties:

- **Normalisation:** $f(0) = 1$.
- **Small-x expansion** [CZ below Eq. (37)]:

$$f(x) = 1 - \frac{x}{6} + \frac{x^2}{60} - \frac{x^3}{840} + \mathcal{O}(x^4)\,. \tag{27}$$

  The general coefficient is $\frac{(-1)^n}{n!} B(n+1, n+1)$ where B is the Beta function.

- **Large-x expansion** [CZ below Eq. (39)]:

$$f(x) = \frac{2}{x} + \frac{4}{x^2} + \mathcal{O}\left(\frac{1}{x^3}\right)\,, \qquad x \to +\infty\,. \tag{28}$$

- **Saddle point:** For $x \to +\infty$, $f(x) \sim \sqrt{\pi/x}\, e^{-x/4}$ (the saddle at $\xi = 1/2$ gives the exponential tail).
- **Useful identity** [CZ Sec. 4.2]:

$$\frac{1}{2}\int_0^1 \mathrm{d}\xi\, e^{-\xi(1-\xi)x}(1-2\xi)^2 = -\frac{1}{x}[f(x) - 1]\,. \tag{29}$$

3 Form Factor Construction: Heat Kernel

3.1 The five BV form factors

[CZ Eqs. (37)–(38); BV (1987) Eqs. (3.44)–(3.48)]

[exact from lit.] In the "Ricci basis" $\{R_{\mu\nu}, R, \mathbf{U}, \Omega_{\mu\nu}\}$ used by CZ, the nonlocal heat trace at $\mathcal{O}(\mathcal{R}^2)$ is:

$$\mathrm{Tr}\, e^{-s\Delta} = \frac{1}{(4\pi s)^{d/2}} \int \mathrm{d}^d x \sqrt{g}\, \mathrm{tr}\Big\{\mathrm{Id} - s\mathbf{U} + s\mathrm{Id}\frac{R}{6} + s^2[\text{form factor terms}] + \mathcal{O}(\mathcal{R}^3)\Big\} \tag{30}$$

with form factor terms:

$$\mathrm{Id}\, R_{\mu\nu}\, f_{\mathrm{Ric}}(s\Box)\, R^{\mu\nu} + \mathrm{Id}\, R\, f_R(s\Box)\, R + R\, f_{R\mathbf{U}}(s\Box)\, \mathbf{U} + \mathbf{U}\, f_{\mathbf{U}}(s\Box)\, \mathbf{U} + \Omega_{\mu\nu}\, f_\Omega(s\Box)\, \Omega^{\mu\nu}\,. \tag{31}$$

The form factors in terms of the master function $f(x)$:

$$\boxed{\begin{aligned} f_{\mathrm{Ric}}(x) &= \frac{1}{6x} + \frac{1}{x^2}[f(x) - 1] \\ f_R(x) &= \frac{1}{32} f(x) + \frac{1}{8x} f(x) - \frac{7}{48x} - \frac{1}{8x^2}[f(x) - 1] \\ f_{R\mathbf{U}}(x) &= -\frac{1}{4} f(x) - \frac{1}{2x}[f(x) - 1] \\ f_{\mathbf{U}}(x) &= \frac{1}{2} f(x) \\ f_\Omega(x) &= -\frac{1}{2x}[f(x) - 1] \end{aligned}} \tag{32}$$

[CZ Eq. (37), confirmed to match BV (1987, 1990) via basis change in CZ App. A]

3.2 Conversion to BV basis

The BV papers use $\hat{P} = E - \frac{R}{6}\mathrm{Id}$ instead of $\mathbf{U}$, with $\mathbf{U} = -E$ (Convention 1.10). The conversion [CZ Eq. (50)] is:

$$\boxed{\begin{aligned} f_1 &= f_{\mathrm{Ric}}\,, \\ f_2 &= f_R + \frac{1}{6} f_{R\mathbf{U}} + \frac{1}{36} f_{\mathbf{U}}\,, \\ f_3 &= -f_{R\mathbf{U}} - \frac{1}{3} f_{\mathbf{U}}\,, \\ f_4 &= f_{\mathbf{U}}\,, \\ f_5 &= f_\Omega\,. \end{aligned}} \tag{33}$$

Substituting (32) into (33) gives the explicit BV form factors. This can be verified against BV (1987) Table 1.

3.3 Conversion to Weyl basis

[CZ Eqs. (41)–(42); Barvinsky–Mirzabekian–Zhytnikov gr-qc/9510037]

For the Weyl basis $\{C_{\mu\nu\rho\sigma}, R, \hat{P}, \Omega_{\mu\nu}\}$:

$$\boxed{\begin{aligned} f_C(x) &= \frac{d-2}{4(d-3)}\, f_{\mathrm{Ric}}(x) \qquad (\text{in } d=4\text{: } f_C = \tfrac{1}{2} f_{\mathrm{Ric}}) \\ f_{R,\mathrm{bis}}(x) &= \frac{d}{4(d-1)}\, f_{\mathrm{Ric}}(x) + f_R(x) \qquad (\text{in } d=4\text{: } f_{R,\mathrm{bis}} = \tfrac{1}{3} f_{\mathrm{Ric}} + f_R) \end{aligned}} \tag{34}$$

The heat trace in the Weyl basis [CZ Eq. (40)]:

$$\mathrm{Tr}\, e^{-s\Delta} = \frac{1}{(4\pi s)^{d/2}} \int \mathrm{d}^d x\, \sqrt{g}\, \mathrm{tr}\Big\{ \mathrm{Id} - s\mathbf{U} + s\frac{R}{6}\mathrm{Id} + s^2 \Big[\mathrm{Id}\, C_{\mu\nu\rho\sigma}\, f_C\, C^{\mu\nu\rho\sigma} + \mathrm{Id}\, R\, f_{R,\mathrm{bis}}\, R + \cdots \Big] \Big\} \tag{35}$$

where $\cdots$ contains $\hat{P}$ and Ω terms unchanged.

Remark 3.1 (Universality). In the Ricci basis, the form factors are **universal** (independent of d apart from the overall $(4\pi s)^{-d/2}$ prefactor). The Weyl basis spoils universality by introducing explicit d-dependent prefactors.

3.4 Short-time (local) limits

[CZ Eq. (38)]

Setting $x = 0$ in (32):

$$\begin{aligned} f_{\mathrm{Ric}}(0) &= \frac{1}{60}\,, \\ f_R(0) &= \frac{1}{120}\,, \\ f_{R\mathbf{U}}(0) &= -\frac{1}{6}\,, \\ f_{\mathbf{U}}(0) &= \frac{1}{2}\,, \\ f_{\Omega}(0) &= \frac{1}{12}\,. \end{aligned} \tag{36}$$

These values reproduce the Seeley–DeWitt coefficient a_4 when inserted into the heat trace.

Remark 3.2 (Cross-check with a_4). The local limit of (30) must reproduce the standard a_4 coefficient [Vassilevich Eq. (4.1)]:

$$a_4 = \frac{1}{360} \int \mathrm{d}^d x\, \sqrt{g}\, \mathrm{tr}\Big[60\, RE + 180\, E^2 + 60\, \Box E + (5R^2 - 2R_{\mu\nu}R^{\mu\nu} + 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})\mathrm{Id} + 30\, \Omega_{\mu\nu}\Omega^{\mu\nu} \Big]\,. \tag{37}$$

This is identical to (60) in Appendix **??**. Note: Vassilevich uses $\Delta = -(g^{\mu\nu}\nabla_\mu\nabla_\nu + E)$ which is the **same** as our convention (10), so his E is the same as our E. The CZ convention has $\mathbf{U} = -E$, so the signs of the E terms need careful tracking when converting. The signs $+60\Box E$ and $+30\,\Omega\Omega$ in Vassilevich's convention yield $-15 R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ for the Dirac operator since $\mathrm{tr}(\Omega^{\mathrm{spin}}\Omega^{\mathrm{spin}}) = -\frac{1}{2}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$.

3.5 Late-time (IR) limits

[CZ Eq. (39)]

$$
\begin{aligned}
f_{\mathrm{Ric}}(x) &= \frac{1}{6x} - \frac{1}{x^2} + \mathcal{O}(x^{-3})\,, \\
f_R(x) &= -\frac{1}{12x} + \frac{1}{2x^2} + \mathcal{O}(x^{-3})\,, \\
f_{R\mathbf{U}}(x) &= -\frac{2}{x^2} + \mathcal{O}(x^{-3})\,, \\
f_{\mathbf{U}}(x) &= \frac{1}{x} + \frac{2}{x^2} + \mathcal{O}(x^{-3})\,, \\
f_{\Omega}(x) &= \frac{1}{2x} - \frac{1}{2x^2} + \mathcal{O}(x^{-3})\,.
\end{aligned}
\tag{38}
$$

These control the UV behavior of the effective action (large $\Box$).

4 BV Nonlocal Effective Action

4.1 From heat kernel to effective action

[BV (1987) Sec. 4; CZ Sec. 6]

The one-loop effective action for a field with Laplace-type operator Δ and mass m:

$$\Gamma^{(1)} = \frac{1}{2}\int_0^\infty \frac{\mathrm{d}s}{s}\, e^{-sm^2}\ \mathrm{Tr}\ e^{-s\Delta} \tag{39}$$

(sign for bosons; the mass term provides IR regularisation).

Substituting the nonlocal heat trace (30) and performing the s-integration converts the heat kernel form factors $f_i(s\Box)$ into effective action form factors $\mathcal{F}_i(\Box/m^2)$:

$$\mathcal{F}_i(\Box/m^2) = \int_0^\infty \frac{\mathrm{d}s}{s}\, e^{-sm^2}\,(4\pi s)^{-d/2}\, s^2\, f_i(s\Box)\,. \tag{40}$$

Remark 4.1. The factor s^2 comes from the fact that the form factor terms in (30) carry an explicit s^2 prefactor at $\mathcal{O}(\mathcal{R}^2)$. The $1/s$ from (39) combines with this to give s.

4.2 Canonical form of the nonlocal effective action

[BV (1990) Eq. (1.1); Barvinsky–Gusev–Vilkovisky–Zhytnikov hep-th/9404187]

At $\mathcal{O}(\mathcal{R}^2)$, the one-loop effective action takes the canonical form in the Weyl basis:

$$\boxed{\Gamma^{(1)}_{\mathrm{NL}} = \frac{1}{2(4\pi)^2}\int \mathrm{d}^4x\,\sqrt{g}\,\Big[C_{\mu\nu\rho\sigma}\,\mathcal{F}_W(\Box/\mu^2)\,C^{\mu\nu\rho\sigma} + R\,\mathcal{F}_R(\Box/\mu^2)\,R\Big]} \tag{41}$$

(dropping the topological Gauss–Bonnet term in $d=4$, and the $\hat{P}$ and Ω terms which are absorbed into the Weyl/Ricci structures for specific field content).

4.3 Massless limit: logarithmic form factors

[BV (1990) Sec. 6; Barvinsky–Wachowski 2306.03780 Sec. 2]

In the massless limit, the effective action form factors develop logarithmic behaviour:

$$\mathcal{F}_i(\Box/\mu^2) \to \beta_i\,\ln(-\Box/\mu^2) + \text{finite local terms}\,. \tag{42}$$

The coefficients β_i are scheme-independent and related to the conformal anomaly. This gives:

$$\boxed{\Gamma^{(1)}_{\mathrm{NL,massless}} = \frac{1}{32\pi^2}\int \mathrm{d}^4x\,\sqrt{g}\,\Big[\beta_W\,C_{\mu\nu\rho\sigma}\,\ln(-\Box/\mu^2)\,C^{\mu\nu\rho\sigma} + \beta_R\,R\,\ln(-\Box/\mu^2)\,R\Big]} \tag{43}$$

Remark 4.2. For all conformally invariant fields, $\beta_R = 0$. This is the deep connection between conformal symmetry and the structure of the nonlocal effective action.

Field type	β_W	β_R
Conformal scalar	1/120	0
Dirac fermion	1/20	0
Gauge vector	1/10	0
Graviton	7/20	0

Table 1: β-coefficients for conformally invariant fields. [BV (1990); Barvinsky–Vilkovisky–Zhytnikov hep-th/9404187]

4.4 The key formula: spectral action to form factors

[GAP] **This is the central gap that NT-1 must bridge.**

The SCT spectral action $\mathrm{Tr}(f(D^2/\Lambda^2))$ gives a different weight than the one-loop effective action $\frac{1}{2}\int \frac{\mathrm{d}s}{s} e^{-sm^2}\,\mathrm{Tr}\; e^{-sD^2}$. Specifically:

$$\mathrm{Tr}\Big(f(D^2/\Lambda^2)\Big) = \int_0^\infty \tilde{f}(t)\;\mathrm{Tr}\Big(e^{-tD^2/\Lambda^2}\Big)\mathrm{d}t\,. \tag{44}$$

Using (24), the $\mathcal{O}(\mathcal{R}^2)$ contribution to the spectral action is:

$$\mathrm{Tr}\Big(f(D^2/\Lambda^2)\Big)\Big|_{\mathcal{R}^2} = \int_0^\infty \tilde{f}(t)\,\frac{1}{(4\pi t/\Lambda^2)^2}\int \mathrm{d}^4x\sqrt{g}\;\mathrm{tr}\Big[\cdots\, s^2 f_i(s\Box)\,\cdots\Big]_{s=t/\Lambda^2}\mathrm{d}t \tag{45}$$

where the form factors f_i are those of (32).

The SCT form factors $F_1(\Box/\Lambda^2)$ and $F_2(\Box/\Lambda^2)$ are then obtained by performing the t-integration against the spectral weight $\tilde{f}(t)$:

$$\boxed{F_i^{\mathrm{SCT}}(\Box/\Lambda^2) = \frac{\Lambda^4}{(4\pi)^2}\int_0^\infty \mathrm{d}t\,\tilde{f}(t)\,(t/\Lambda^2)^{2-d/2}\,f_i(t\Box/\Lambda^2)} \tag{46}$$

where f_i are the heat kernel form factors from (32) or (34).

Remark 4.3 (Notation for the target). In the Weyl basis, the final result is written as:

$$S_{\mathrm{SCT}}\Big|_{\mathcal{R}^2} = \int \mathrm{d}^4x\,\sqrt{g}\,\Big[F_1(\Box/\Lambda^2)\,C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + F_2(\Box/\Lambda^2)\,R^2\Big]\,, \tag{47}$$

where $F_1 \propto f_C$ integrated against $\tilde{f}$, and $F_2 \propto f_{R,\mathrm{bis}}$ integrated against $\tilde{f}$ (plus contributions from $\hat{P}$ and Ω terms traced for the Dirac operator).

5 Computational Recipe

The step-by-step procedure for computing $F_1(\Box/\Lambda^2)$ and $F_2(\Box/\Lambda^2)$ from the SCT spectral action.

5.1 Step 1: Identify the operator

The SCT Dirac operator gives $D^2 = \Delta$ with:

- Endomorphism: $E = -\frac{R}{4}\mathrm{Id}_4 \implies \mathbf{U}_{\mathrm{CZ}} = -E = \frac{R}{4}\mathrm{Id}_4$
- Bundle curvature: $\Omega_{\mu\nu} = \frac{1}{4}R_{\mu\nu\rho\sigma}\gamma^\rho\gamma^\sigma$
- BV endomorphism: $\hat{P} = E - \frac{R}{6}\mathrm{Id}_4 = -\frac{5R}{12}\mathrm{Id}_4$
- Spin dimension: $N = \mathrm{tr}\,\mathrm{Id}_4 = 4$

5.2 Step 2: Write the nonlocal heat trace for D^2

Using (24) with the Dirac data above. The CZ form factors $f_1, \ldots, f_5$ act on the CZ endomorphism $\mathbf{U}_{\mathrm{CZ}} = -E = \frac{R}{4}\mathrm{Id}_4$, **not** on $\hat{P}$. Writing the heat trace in CZ notation:

$$\begin{aligned}\mathrm{Tr}\ e^{-sD^2} = \frac{1}{(4\pi s)^2}\int \mathrm{d}^4x\,\sqrt{g}\,\Big\{4 - s\,\frac{5R}{3} \\ + s^2\Big[4\,R_{\mu\nu}\,f_1(s\Box)\,R^{\mu\nu} + 4\,R\,f_2(s\Box)\,R \\ + R\,f_3(s\Box)\ \mathrm{tr}(\mathbf{U}_{\mathrm{CZ}}) + f_4(s\Box)\ \mathrm{tr}(\mathbf{U}_{\mathrm{CZ}}^2) + f_5(s\Box)\ \mathrm{tr}(\Omega\Omega)\Big] + \mathcal{O}(\mathcal{R}^3)\Big\}\end{aligned} \tag{48}$$

After taking the trace over spinor indices (with $E = -R/4$, $\mathbf{U}_{\mathrm{CZ}} = R/4$, $\hat{P} = -5R/12$):

- $\mathrm{tr}(\mathbf{U}_{\mathrm{CZ}}) = 4 \cdot R/4 = R$
- $\mathrm{tr}(\mathbf{U}_{\mathrm{CZ}}^2) = 4 \cdot R^2/16 = R^2/4$
- $\mathrm{tr}(\hat{P}) = 4 \cdot (-5R/12) = -5R/3$
- $\mathrm{tr}(\hat{P}^2) = 4 \cdot 25R^2/144 = 25R^2/36$
- $\mathrm{tr}(\Omega_{\mu\nu}\Omega^{\mu\nu}) = -\frac{1}{2}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$

So the $\mathcal{O}(\mathcal{R}^2)$ part becomes:

$$\begin{aligned}\mathrm{Tr}\ e^{-sD^2}\Big|_{\mathcal{R}^2} = \frac{s^2}{(4\pi s)^2}\int \mathrm{d}^4x\,\sqrt{g}\,\Big[4\,R_{\mu\nu}\,f_1(s\Box)\,R^{\mu\nu} \\ + \Big(4f_2 + f_3 + \frac{1}{4}f_4\Big)\,R\,(s\Box)\,R - \frac{1}{2}\,R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\,f_5(s\Box)\Big]\end{aligned} \tag{49}$$

5.3 Step 3: Convert to Weyl basis

Using the Gauss–Bonnet identity at $\mathcal{O}(\mathcal{R}^2)$ [CZ App. A; Avramidi (2000)]:

$$\int \mathrm{d}^4x\,\sqrt{g}\,R_{\mu\nu\rho\sigma}\,h(\Box)\,R^{\mu\nu\rho\sigma} = \int \mathrm{d}^4x\,\sqrt{g}\left[4\,R_{\mu\nu}\,h(\Box)\,R^{\mu\nu} - R\,h(\Box)\,R\right] + \mathcal{O}(\mathcal{R}^3)\,. \tag{50}$$

This means the $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ term from $\mathrm{tr}(\Omega\Omega)$ can be absorbed into $R_{\mu\nu}R^{\mu\nu}$ and R^2 terms.

Then convert $R_{\mu\nu}R^{\mu\nu} \to C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \frac{2}{d-2}R_{\mu\nu}R^{\mu\nu}_{\text{tracefree}} + \ldots$ using (6).

The final result should take the form:

$$\mathrm{Tr}\;e^{-sD^2}\Big|_{\mathcal{R}^2} = \frac{s^2}{(4\pi s)^2}\int \mathrm{d}^4x\,\sqrt{g}\left[h_C(s\Box)\,C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + h_R(s\Box)\,R^2\right] \tag{51}$$

where h_C and h_R are combinations of the f_i with the appropriate Dirac trace coefficients.

5.4 Step 4: Integrate against spectral weight

The SCT form factors are:

$$\boxed{\begin{aligned} F_1(\Box/\Lambda^2) &= \frac{\Lambda^4}{(4\pi)^2}\int_0^\infty \mathrm{d}t\,\tilde{f}(t)\,h_C(t\Box/\Lambda^2) \\ F_2(\Box/\Lambda^2) &= \frac{\Lambda^4}{(4\pi)^2}\int_0^\infty \mathrm{d}t\,\tilde{f}(t)\,h_R(t\Box/\Lambda^2) \end{aligned}} \tag{52}$$

where the t-integration converts heat kernel form factors to spectral action form factors.

Remark 5.1 (Difference from one-loop effective action). In the standard one-loop computation, the weight is $\frac{1}{s}e^{-sm^2}$ (gives logarithmic form factors in massless limit). In SCT, the weight is $\tilde{f}(t)$ (the inverse Laplace transform of the spectral function f), which provides UV regularisation at scale Λ. The SCT form factors are therefore **manifestly finite** — no renormalisation needed.

5.5 Step 5: Check limiting cases

1. **Local limit** ($\Box \to 0$): $F_1(0)$ and $F_2(0)$ should reproduce the c_1 and c_2 coefficients from the Seeley–DeWitt computation (Derivation 1 in `SCT_derivations.tex`).

2. **Flat space limit** ($R \to 0$): Both F_1 and F_2 should vanish trivially.

3. **Large-$\Box$ limit** ($\Box/\Lambda^2 \gg 1$): Form factors should decay (UV behaviour controlled by the spectral function f).

4. **Comparison with β-coefficients**: In the limit where f approaches a sharp cutoff, the form factors should reproduce the logarithmic running with the correct $\beta_W = 1/20$ for the Dirac field.

6 Key Formulas from Literature

6.1 Codello–Zanusso diagrammatic computation

[arXiv:1203.2034, Secs. 3–5]

The CZ paper provides an independent re-derivation of the BV form factors using a diagrammatic expansion of the heat kernel. Key results:

1. **Perturbative expansion of heat kernel** [CZ Eq. (16)]:

$$K^s_{xy} = K^s_{0,xy} - s\int_0^1 \mathrm{d}t \int_z K^{s(1-t)}_{0,xz} V_z K^{st}_{0,zy} + \mathcal{O}(V^2) \tag{53}$$

   where $K^s_{0,xy} = (4\pi s)^{-d/2} e^{-(x-y)^2/4s}$ is the flat space heat kernel, and V contains all interactions (metric fluctuation $h_{\mu\nu}$, connection A_μ, and endomorphism $\mathbf{U}$).

2. **General n-th order term** [CZ Eq. (17)]:

$$(-s)^n \int_0^1 \mathrm{d}t_1 \cdots \int_0^{t_{n-1}} \mathrm{d}t_n \int_{z_1\cdots z_n} K^{s(1-t_1)}_{0,xz_1} V_{z_1} \cdots V_{z_n} K^{st_n}_{0,z_n y} \tag{54}$$

3. **Master function from momentum integral** [CZ Eq. (36)]:

   The basic form factor emerges from the sunset diagram:

$$f(sp^2) = \int_0^1 \mathrm{d}\xi\, e^{-\xi(1-\xi)sp^2} \tag{55}$$

   where $\xi = t_1 - t_2$ is a reparametrisation of the proper-time integration variables.

4. **All five form factors confirmed** [CZ Sec. 5]: The CZ computation reproduces the BV form factors in (32) by explicit diagrammatic evaluation of 1-point and 2-point functions of the heat trace.

6.2 Avramidi generating function method

[Avramidi, hep-th/9704166 Secs. 2–4; Phys. Lett. B **238** (1990) 92]

1. **Covariantly constant curvature:** Avramidi obtains closed-form heat kernel for covariantly constant backgrounds ($\nabla_\mu R_{\nu\rho\sigma\tau} = 0$, $\nabla_\mu \Omega_{\nu\rho} = 0$, $\nabla_\mu E = 0$) using a Lie-algebraic method. The heat kernel diagonal is expressed as an average over a Lie group with Gaussian measure [gr-qc/9408028, hep-th/9602169].

2. **Partial summation:** By expanding around the covariantly constant case and keeping all orders in $s\Box$ while truncating at second order in curvature, Avramidi obtains the same form factor structure as BV, confirming the five universal form factors $f_1, \ldots, f_5$.

3. **Off-diagonal heat kernel:** [CZ App. B; Avramidi (2000) Ch. 9]

   The off-diagonal heat kernel (before taking the coincidence limit) has additional form factors $g_{\mathbf{U}}(x)$ and $g_R(x)$:

$$K^s(x,x) = \frac{\sqrt{g}}{(4\pi s)^{d/2}} \Big\{ \mathrm{Id} + s\, g_{\mathbf{U}}(s\Box)\, \mathbf{U} + s\, \mathrm{Id}\, g_R(s\Box)\, R + \mathcal{O}(\mathcal{R}^2) \Big\} \tag{56}$$

with [CZ App. B]:

$$g_{\mathbf{U}}(x) = -f(x)\,, \qquad g_R(x) = -\frac{1}{2x} + \frac{1}{4x}(x+2)f(x)\,. \tag{57}$$

These satisfy $g_{\mathbf{U}}(0) = -1$ and $g_R(0) = 1/6$.

6.3 Vassilevich: Seeley–DeWitt coefficients

[Vassilevich, hep-th/0306138, Sec. 4]

The standard Seeley–DeWitt coefficients for a Laplace-type operator $\Delta = -(g^{\mu\nu}\nabla_\mu\nabla_\nu + E)$ on a closed manifold:

$$a_0 = \frac{1}{(4\pi)^{d/2}} \int \mathrm{d}^d x\, \sqrt{g}\ \mathrm{tr}\,\mathrm{Id} \tag{58}$$

$$a_2 = \frac{1}{(4\pi)^{d/2}} \frac{1}{6} \int \mathrm{d}^d x\, \sqrt{g}\ \mathrm{tr}\Big(6E + R\mathrm{Id}\Big) \tag{59}$$

$$\begin{aligned} a_4 = \frac{1}{(4\pi)^{d/2}} \frac{1}{360} \int \mathrm{d}^d x\, \sqrt{g}\ \mathrm{tr}\Big[&60\, RE + 180\, E^2 + 60\,\Box E \\ &+ (5R^2 - 2R_{\mu\nu}R^{\mu\nu} + 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})\mathrm{Id} + 30\,\Omega_{\mu\nu}\Omega^{\mu\nu}\Big] \end{aligned} \tag{60}$$

Remark 6.1 (Sign in a_4). [needs sign check] Vassilevich Eq. (4.1) writes a_4 with $+180E^2$ and $+30\,\Omega\Omega$. Some references (e.g., Gilkey) write $-180E^2$ and $-30\,\Omega\Omega$ due to a different sign convention for E. Our $E_{\text{Vassilevich}} = E_{\text{ours}}$ (both use $\Delta = -\nabla^2 - E$), so (60) applies directly. However, the sign of the $\Omega\Omega$ term depends on the convention for the bundle curvature: if $\Omega_{\mu\nu} = [D_\mu, D_\nu]$ (our convention, same as CZ and Vassilevich), then $\mathrm{tr}(\Omega\Omega)$ for the spin connection is **negative**: $\mathrm{tr}(\Omega^{\text{spin}}\Omega^{\text{spin}}) = -\frac{1}{2}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$. So $+30 \times (-\frac{1}{2}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}) = -15R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ enters a_4.

6.4 Iochum–Levy–Vassilevich: Spectral action beyond weak field

[arXiv:1108.3749; arXiv:1201.6637]

ILV compute the spectral action $\mathrm{Tr}\ f(D^2/\Lambda^2)$ covariantly to second order in the gauge connection, going beyond the weak-field (Seeley–DeWitt) approximation. Key results:

1. They confirm that the covariant perturbation theory (BV technique) applies to the spectral action: the form factors are the same as those in the heat kernel, just integrated against a different weight.

2. For a commutative spectral triple on a torus, they compute the spectral action to second order in A_μ and find [1108.3749 abstract]: in the UV regime $p \to \infty$, the action decays as $1/p^4$ in any even dimension.

3. The convergence of the nonlocal expansion is controlled by the smoothness of f: for Schwartz-class f, the expansion converges for all $\Box$.

[GAP] **The explicit form of the spectral action form factors (as opposed to the heat kernel form factors) for a general spectral function f and the full Dirac operator on a curved manifold has not been computed in the literature.** This is precisely the gap NT-1 aims to fill.

6.5 Barvinsky–Vilkovisky: parametric integral representation

[BV (1987) Eq. (3.44) ff.; BV (1990) Eqs. (3.1)–(3.4)]

The BV form factors have the parametric integral representation:

$$f_i(x) = \int_0^1 \mathrm{d}\xi\, \Phi_i(\xi)\, e^{-\xi(1-\xi)x} \tag{61}$$

where the $\Phi_i(\xi)$ are polynomial weight functions. For the five form factors at $\mathcal{O}(\mathcal{R}^2)$:

$$\boxed{\begin{aligned} \Phi_1(\xi) &= \frac{1}{6} + (1 - 6\xi + 6\xi^2)\frac{1}{[\xi(1-\xi)]^2} \cdot (\text{subtracted}) \\ \Phi_4(\xi) &= \frac{1}{2} \\ \Phi_5(\xi) &= \frac{1}{2} \cdot \frac{(1-2\xi)^2}{[\text{from identity (29)}]} \end{aligned}} \tag{62}$$

[GAP] **The exact polynomial weight functions Φ_1, Φ_2, Φ_3 from BV (1987) are needed.** The forms given above are incomplete. The original BV papers (Nucl. Phys. B **282** and **333**) should be consulted for the precise expressions. The arXiv paper 0911.1168 (BV IV) contains the complete results through third order in curvature.

Remark 6.2. The parametric representation (61) is particularly useful for the NT-1 computation because it allows the s-integration (Step 4 in Sec. 5) to be performed by interchanging the order of integration:

$$F_i^{\mathrm{SCT}} \propto \int_0^1 \mathrm{d}\xi\, \Phi_i(\xi) \int_0^\infty \mathrm{d}t\, \tilde{f}(t)\, e^{-\xi(1-\xi)t\Box/\Lambda^2}\,. \tag{63}$$

The inner integral is then a function of $\xi(1-\xi)\Box/\Lambda^2$ determined by the spectral function f.

7 Known Results for Cross-Checks

7.1 Seeley–DeWitt limit

In the local limit $\Box/\Lambda^2 \to 0$, the form factors F_1 and F_2 should reduce to the curvature-squared coefficients obtained from the a_4 Seeley–DeWitt coefficient.

From `SCT_derivations.tex` Eq. (1.22):

$$a_4(D^2) = \frac{1}{16\pi^2 \cdot 360} \int_M \Big[12\,\Box R + 5R^2 - 8R_{\mu\nu}R^{\mu\nu} - 7R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\Big]\sqrt{g}\,\mathrm{d}^4x \tag{64}$$

In the Weyl basis: $5R^2 - 8R_{\mu\nu}R^{\mu\nu} - 7R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 11\,E_4 - 18\,C^2$.

So the local limit of the spectral action at $\mathcal{O}(\mathcal{R}^2)$ gives:

$$F_1(0) = -\frac{18\,f_4}{5760\pi^2} = -\frac{f_4}{320\pi^2}\,, \qquad F_2(0) = 0\,, \tag{65}$$

where $f_4 = f(0)$ is the zeroth moment of the spectral function. The vanishing $F_2(0) = 0$ is a consequence of conformal invariance of the massless Dirac field: $\beta_R = 0$ (verified independently in NT1_form_factors, Sec. 5).

Remark 7.1. The E_4 (topological) part does not contribute to F_1 or F_2 in the equations of motion, but enters the action. When checking the local limit, one should compare with the full a_4 including E_4. The $\Box R$ term integrates to zero on a closed manifold.

7.2 Conformal flatness limit

On a conformally flat manifold ($C_{\mu\nu\rho\sigma} = 0$), only the $F_2\,R^2$ term survives. This should be consistent with the known spectral action on spaces like S^4.

7.3 Weak-field flat-space limit

Expanding around flat space $g_{\mu\nu} = \delta_{\mu\nu} + h_{\mu\nu}$, the form factors in momentum space should reproduce the 2-point functions computed by CZ:

$$\left.\frac{\delta^2\,\mathrm{Tr}\,K^s}{\delta h_{p;\mu\nu}\delta h_{-p;\alpha\beta}}\right|_{\Phi=0} = \frac{1}{(4\pi s)^{d/2}}\Big\{\text{known tensor structure from CZ (32)}\Big\} \tag{66}$$

7.4 Two-dimensional limit

In $d = 2$, the heat kernel expansion terminates at second order in curvature [CZ Sec. 5.1; BV (1990)]. The combined form factor is [CZ Eq. (43)]:

$$f_{R,2d}(x) = \frac{1}{32}f(x) + \frac{1}{16x}[2f(x) - 1] + \frac{3}{8x^2}[f(x) - 1]\,. \tag{67}$$

This has the late-time expansion $f_{R,2d}(x) = 2/x^3 + \mathcal{O}(x^{-4})$, with both x^{-1} and x^{-2} coefficients vanishing — a nontrivial check.

7.5 Comparison with logarithmic running

For a sharp cutoff spectral function (step function), the spectral action form factors should reduce in the UV to:

$$F_i(\Box/\Lambda^2) \xrightarrow{\Box/\Lambda^2 \to 0} \beta_i \, \ln(\Lambda^2/\Box) + \text{const}\,, \tag{68}$$

matching the standard running with the correct β-coefficients from Table 1.

For the Dirac field: $\beta_W = 1/20$. This provides a critical cross-check of the final result.

7.6 UV decay

[ILV 1108.3749]

For a smooth spectral function f (Schwartz class), the spectral action should decay in the UV ($p \to \infty$) as $1/p^4$ in $d = 4$:

$$F_i(p^2/\Lambda^2) \sim \frac{1}{p^4}\,, \qquad p^2/\Lambda^2 \to \infty\,. \tag{69}$$

This is a consequence of the smoothness of f and is specific to the spectral action (the one-loop effective action has logarithmic growth instead).

Summary of Gaps

The following items are **not available** in the extracted literature and must be computed in the NT-1 derivation:

1. **Explicit Dirac trace reduction:** The spinor trace of the five form factor terms in (49), converting from the $\{R_{\mu\nu}, R, \hat{P}, \Omega_{\mu\nu}\}$ basis to the Weyl basis $\{C_{\mu\nu\rho\sigma}, R\}$ with all Dirac-specific coefficients computed.

2. **The s-integration against the spectral weight:** The explicit integral (52) converting heat kernel form factors h_C, h_R into spectral action form factors F_1, F_2 for a general spectral function f. *This is the core new computation.*

3. **BV parametric weights Φ_1, Φ_2, Φ_3:** The exact polynomial weight functions in the parametric representation (61) for f_1, f_2, f_3. These are available in BV (1987) Nucl. Phys. B **282** but were not extracted here.

4. **Closed-form result for specific f:** For common choices of spectral function (e.g., $f(u) = e^{-u}$, or $f(u) = \chi_{[0,1]}(u)$), the integrals in (52) can be evaluated in closed form. These are not in the literature.

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